

Spenso

Typed tensors, symbolic structures, and executable tensor networks

1 Overview

Spenso represents and evaluates tensors with abstract indices. It separates a tensor's structure—representations, slots, dimensions, names, and index order—from its stored data. Dense and sparse storage, symbolic and parametric values, automatic index matching, and network execution build on that separation.

Two questions, two layers

Tensor structure answers “which indices and representations does this object carry?” Tensor data answers “which values occupy that structure?” Contraction first matches compatible slots, then combines data. Keep these phases distinct when diagnosing a shape, duality, or contraction error.

1.1 From tensors to networks

The core package supports three related levels of work:

- tensor structures describe slots and abstract indices;
- dense, sparse, numeric, symbolic, and parametric tensors attach data to those structures;
- tensor networks represent a collection of tensors and choose an execution or contraction strategy.

Networks can be assembled directly. With the shadowing feature, they can also be inferred from Symbolica expressions whose functions carry recognizable tensor structure. Parsing is not mere string conversion: representation initialization, index variance, names, and the tensor library determine the inferred network.

```
[dependencies]  
spenso = "0.6"
```

Enable shadowing only when Symbolica-backed parsing or symbolic operations are needed. It pulls in the Symbolica integration and corresponding Linnet support. The default core remains useful for typed tensors and contractions without that layer.

1.2 Companion packages

Component ownership

spenso owns the tensor model and execution engine. spenso-macros owns the SimpleRepresentation derive used to declare custom index representations. spenso-hep-lib owns concrete high-energy-physics tensor-library data such as gamma matrices and projectors. The spynso3 adapter owns the Python community module. Their versions and feature gates are independent, so select and upgrade each component explicitly.

Spenso uses Linnet for the underlying network graph. Idenso is the symbolic-identity layer for Spenso-formatted Symbolica expressions. GammaLoop consumes these components inside a broader collider workflow.

The Spenso README gives a compact package summary. The Rust and Python API sections list the public modules, traits, feature gates, and import paths.

2 Tutorial

This tutorial performs one concrete tensor contraction in Rust. The example makes the tensor structure, dual index matching, storage, and numerical result visible—the same layers that a larger Spenso network coordinates automatically.

2.1 Prerequisites

Use Rust 1.85 or newer (Spenso uses Rust 2024 edition), then create a binary project:

```
cargo new spenso-first-contraction
cd spenso-first-contraction
cargo add spenso@0.6
```

This first contraction uses dense integer tensors and needs neither Symbolica nor the shadowing feature.

2.2 Contract one shared index

Replace `src/main.rs` with:

```
use spenso::{
    contraction::Contract,
    structure::{
        OrderedStructure, PermutedStructure,
        representation::{LibraryRep, Lorentz, RepName},
        slot::{DualSlotTo, IsAbstractSlot},
    },
    tensors::data::{DenseTensor, GetTensorData, SetTensorData},
};

fn main() {
    let rep = Lorentz {};
    let left_free = rep.new_slot(2, 0).to_lib();
    let right_free = rep.new_slot(2, 2).to_lib();
    let shared = rep.new_slot(2, 10).to_lib();

    let left_structure: OrderedStructure<LibraryRep> =
        PermutedStructure::from_iter([left_free, shared]).structure;
    let right_structure: OrderedStructure<LibraryRep> =
        PermutedStructure::from_iter([right_free, shared.dual()]).structure;

    let mut left = DenseTensor::<i32, _>::zero(left_structure);
    let mut right = DenseTensor::<i32, _>::zero(right_structure);
    left.set(&[0, 1], 2).unwrap();
    right.set(&[0, 1], 5).unwrap();

    let result = left.contract(&right).unwrap();
    assert_eq!(*result.get_ref([0, 0]).unwrap(), 10);
    println!("result structure: {}", result.structure);
    println!("result data: {:?}", result.data);
}
```

Run `cargo run`. Success means the assertion passes and the printed result contains 10 at the component selected by the two remaining free indices. Spenso matched `shared` with `shared.dual()`, summed that axis, and preserved `left_free` and `right_free` in the output structure. The contraction follows index identity and duality, not merely equal dimensions.

Structure is part of the value

The data vectors alone do not say which axes may contract. Preserve the `OrderedStructure` with the tensor,

and inspect the resulting structure before interpreting a flat component offset. A dimension match with unrelated abstract indices produces an exterior product, not the contraction shown above.

2.3 From two tensors to a network

Pairwise Contract is the clearest first success. For a larger expression, place tensors in a Network, inspect its graph, choose or benchmark a contraction strategy, and execute it. Use DataTensor when individual nodes may be dense or sparse. Enable shadowing only when the network must mix concrete data with Symbolica-backed parametric tensors.

2.4 Troubleshooting and next steps

- If contract leaves both shared-looking axes free, print their slots and check that one is the dual of the other; dimensions alone are insufficient.
- If set rejects a component, compare the coordinate count and each coordinate with the structure's rank and dimensions.
- If a sparse contraction reports an empty input, distinguish an intentionally zero tensor from a tensor whose nonzero entries were never populated.
- Continue with the tensor-structures and networks manual before selecting an automatic contraction strategy.

3 Tensor data, contraction, and network execution

Spensor keeps tensor structure and tensor data separate so the same structural operations work with dense, sparse, symbolic, or parametric storage. A structure supplies ordered slots, representations, dimensions, and abstract indices. Data supplies values and a storage-specific access strategy. Contraction first establishes structural compatibility and only then chooses a data operation.

3.1 Dense, sparse, and parametric data

Dense storage is appropriate when most components are populated and a predictable contiguous layout matters. Sparse storage is appropriate when zero structure dominates, but contraction cost depends on index overlap and lookup behavior rather than only the number of stored values. Parametric and symbolic tensors defer numeric substitution; keep their parameter environment with the tensor when serializing or moving a calculation between stages.

Conversions must preserve the structure's slot order. A tensor with the same dimensions but a different representation or variance is not interchangeable until an explicit structural operation establishes that relationship.

3.2 Contraction and planning

Pairwise contraction matches compatible dual slots, determines the output structure, and then contracts the selected data backends. A tensor network generalizes this to many nodes connected through abstract indices. Planning selects a contraction order; execution applies that plan to the current data.

The cheapest-looking local pair is not always the cheapest complete plan. Intermediate tensor dimensions, sparsity, symbolic expression growth, and reusable subexpressions can dominate. Record the planner/configuration with performance results and invalidate a stored plan when the network structure changes.

Plan structure, execute data

A plan is meaningful for the network structure it was derived from. Replacing values without changing slots can reuse it; adding a tensor, changing a representation, or rewiring an index requires validation or replanning.

3.3 Symbolica shadowing

With shadowing, Spenso recognizes registered tensor functions inside Symbolica expressions and builds a network whose nodes shadow those subexpressions. Initialization of representations and the tensor library must happen before parsing. Unknown functions remain symbolic rather than silently acquiring tensor semantics.

Shadowing is useful for extracting a contraction problem while retaining a reversible link to the source expression. Keep the replacement map until the computed result has been substituted back. For rewrite identities and index cooking, use `Idenso` Spenso itself owns storage, contraction, and execution.

3.4 Custom representations and the HEP library

`SimpleRepresentation` derives the repetitive representation plumbing for a user enum. The derive input still defines domain semantics: duality, dimension, slot compatibility, and any Symbolica tags must agree. The resulting representations implement the ordinary Rust traits used by the rest of Spenso; the `spenso-macros` API reference lists the supported derive attributes.

`spenso-hep-lib` registers concrete high-energy-physics structures such as gamma matrices and projectors. Treat that library as shared domain data. `GammaLoop` uses the same definitions.

Graph and tensor ownership

`Linnet` owns connectivity, cuts, cycles, and graph mutation. Spenso owns which connected slots contract and how a plan is executed. Keeping that boundary explicit prevents a graph rewrite from being mistaken for a tensor-algebra identity.

4 Rust, macros, and Python APIs

4.1 Core Rust package

The `spenso` crate organizes its public surface into structure, tensors, contraction, network, iterators, and algebra. The important abstractions form a progression:

- `TensorStructure` and related traits describe slots, names, and contraction compatibility;
- `DenseTensor`, `SparseTensor`, and the heterogeneous tensor enums own storage;
- contraction traits perform pairwise or multi-tensor operations;
- network stores and libraries bind symbolic tensor names to concrete data and execute a graph.

Trait implementations and generic constraints determine which combinations of structure and data support an operation. Consult the Rust API reference when a method is unavailable for a particular tensor type. The shadowing API and symbolic parallelism controls require their corresponding Cargo features.

4.2 Proc macros and HEP data

`spenso-macros` is a separate proc-macro crate. Its `SimpleRepresentation` derive generates the representation and duality boilerplate used by Spenso index types. A declaration supplies a symbolic name and chooses either `self_dual` or a `dual_name`:

```
use spenso_macros::SimpleRepresentation;
use spenso::structure::representation::RepName;

#[derive(SimpleRepresentation)]
#[derive(Debug, Clone, Copy, PartialEq, Eq, Hash, PartialOrd, Ord, Default)]
#[representation(name = "flavor", dual_name = "AntiFlavor")]
struct Flavor {}
```

The proc-macro API reference lists the derive's helper attributes, allowed targets, and feature conditions. Macro expansion happens at compile time and produces ordinary Rust implementations.

`spenso-hep-lib` supplies domain data and tensor-library construction for high-energy physics. It is intentionally separate from the generic core. Users who need gamma matrices or physics projectors add that package; generic Spenso users do not inherit those conventions implicitly.

4.3 Python community module

An adapter, not a Spenso wheel

Python users import `symbolica.community.spenso`. The implementation is the `spynso3` Rust adapter and is distributed through the Symbolica community-module mechanism. Enabling Spenso's Rust python feature only enables conversion interoperability; it does not create a standalone importable `spenso` Python distribution.

For a released Python assembly that includes Spenso, install the provider's Symbolica distribution with `python -m pip install symbolica`, then verify `python -c "import symbolica.community.spenso"`. Module availability follows the Symbolica assembly version; a local Spenso source checkout does not add the module to an installed wheel. Source embedders must add `spynso3` to the external `symbolica-community` assembly and invoke its `SymbolicaCommunityModule` registration while building that extension; building the Rust crate alone does not inject the module into another Symbolica wheel.

```
from symbolica.community.spenso import (
    Representation,
    Tensor,
    TensorIndices,
)

rep = Representation.euc(2)
indices = TensorIndices(rep("i"), rep("j"))
identity = Tensor.dense(indices, [1.0, 0.0, 0.0, 1.0])
```

The Python surface includes tensor and structure types, tensor networks, execution modes, tensor libraries, expression evaluators, and symbolic-parallelism controls. Its signatures can differ from the generic Rust API because `spynso3` provides the Python-specific conversions and defaults.

Source starting points are the [core crate](#), the [derive crate](#), the [HEP library](#), and the [Python adapter](#).

5 Releases and change history

Version coverage

This version of Spenso is 0.6.0, but the published changelog currently ends at 0.5.6. Release notes for 0.6.0 are not yet available, so the history below is incomplete for that version. `spenso-macros`, `spenso-hep-lib`, and `spynso3` have independent versions and changelogs.

When upgrading, inspect the history for tensor structure, contraction, network parsing, Symbolica integration, and Python API or signature changes. Also check each companion component used by the application: a core Spenso release does not imply a release of the macro, HEP, or Python adapter package.

For reproducible upgrades, record the versions of Spenso and every companion component used by the application. Do not assume that changes between 0.5.6 and 0.6.0 are fully represented in the available notes.

Release notes: [Spenso](#), [macros](#), [HEP library](#), and [Python adapter](#).

API reference

The packages available in this version are listed below. Follow an item link for its complete language reference.

spenso

Spensor's supported tensor and network API. The scope is *intentionally* ordered so foundational tensor types precede planning and execution.

Tensor structure

Describes ordered slots, dimensions, representations, and contraction compatibility independently of data.

Describes ordered slots, dimensions, representations, and contraction compatibility independently of data.

```
pub trait TensorStructure
{
  type Slot : IsAbstractSlot + DualSlotTo < Dual = Self :: Slot > ; type
  Indexed : TensorStructure < Indexed = Self :: Indexed, Slot = Self :: Slot
  > ; fn
  reindex(self, indices : & [< Self :: Slot as IsAbstractSlot > :: Aind],)
  -> Result < PermutedStructure < Self :: Indexed > , StructureError > ; fn
  dual(self) -> Self; fn string_rep(& self) -> String
  {
    self.external_structure_iter().map(| s | s.to_string()).collect :: <
    Vec < _ > > ().join("")
  } fn external_structure_iter(& self) -> impl Iterator < Item = Self ::
  Slot > ; fn external_dims_iter(& self) -> impl Iterator < Item = Dimension
  > ; fn external_reps_iter(& self,) -> impl Iterator < Item =
  Representation < < Self :: Slot as IsAbstractSlot > :: R > > ; fn
  external_indices_iter(& self) -> impl Iterator < Item = < Self :: Slot as
  IsAbstractSlot > :: Aind > ; fn
  get_aind(& self, i : impl Into < SlotIndex >) -> Option < < Self :: Slot
  as IsAbstractSlot > :: Aind > ; fn
  get_rep(& self, i : impl Into < SlotIndex > ,) -> Option < Representation
  < < Self :: Slot as IsAbstractSlot > :: R > > ; fn
  get_dim(& self, i : impl Into < SlotIndex >) -> Option < Dimension > ; fn
  get_slot(& self, i : impl Into < SlotIndex >) -> Option < Self :: Slot > ;
  fn order(& self) -> usize;
  #[doc =
  " returns the list of slots that are the external indices of the tensor"]
  fn external_structure(& self) -> Vec < Self :: Slot >
  { self.external_structure_iter().collect() } fn to_shell(self) ->
  TensorShell < Self > where Self : Sized, { TensorShell :: new(self) } fn
  contains_matching(& self, slot : & Self :: Slot) -> bool
  { self.external_structure_iter().any(| s | s.matches(slot)) } fn
  external_reps(& self) -> Vec < Representation < < Self :: Slot as
  IsAbstractSlot > :: R > > { self.external_reps_iter().collect() } fn
  external_indices(& self) -> Vec < < Self :: Slot as IsAbstractSlot > ::
  Aind > { self.external_indices_iter().collect() }
  #[doc =
  " checks if the tensor has the same exact structure as another tensor"] fn
  same_content(& self, other : & Self) -> bool { self.same_external(other) }
  #[doc =
  " Given two [`TensorStructure`]s, returns the index of the first matching slot in
  each external index list, along with a boolean indicating if there is a single
  match"]
```

```

fn match_index(& self, other : & Self) -> Option < (bool, usize, usize) >
{
    let posmap =
    self.external_structure_iter().enumerate().map(| (i, slot) |
    (slot, i)).collect :: < AHashMap < _, _ > > (); let mut first_pair :
    Option < (usize, usize) > = None; for (j, slot) in
    other.external_structure_iter().enumerate()
    {
        if let Some(& i) = posmap.get(& slot.dual())
        {
            if let Some((i, j)) = first_pair
            { return Some((false, i, j)); } first_pair = Some((i, j));
        }
    } first_pair.map(| (i, j) | (true, i, j))
}
#[doc =
" Given two [`TensorStructure`]s, returns the index of the first matching slot in
each external index list"]
fn match_indices(& self, other : & Self) -> Option <
(Permutation, Vec < bool > , Vec < bool >) >
{
    let mut self_matches = vec! [false; self.order()]; let mut perm = Vec
:: new(); let mut other_matches = vec! [false; other.order()]; let
posmap =
self.external_structure_iter().enumerate().map(| (i, slot) |
(slot, i)).collect :: < AHashMap < _, _ > > (); for (j, slot_other) in
other.external_structure_iter().enumerate()
{
    if let Some(& i) = posmap.get(& slot_other.dual())
    {
        self_matches [i] = true; other_matches [j] = true;
        perm.push(i);
    }
} if perm.is_empty() { None } else
{
    let p : Permutation = Permutation :: sort(& perm);
    Some((p, self_matches, other_matches))
}
} #[doc = " Identify the repeated slots in the external index list"] fn
traces(& self) -> Vec < [usize; 2] >
{
    let mut positions : HashMap < & Self as TensorStructure > :: Slot, Vec
< usize > > = HashMap :: new(); for (index, key) in
self.external_structure_iter().enumerate()
{
    if let Some(v) = positions.get_mut(& key.dual())
    { v.push(index); } else { positions.insert(key, vec! [index]); }
}
positions.into_values().filter_map(| indices |
{
    if indices.len() == 2 { Some([indices [0], indices [1]]) } else
    { None }
}).collect()
}
#[doc =
" yields the (outwards facing) shape of the tensor as a list of dimensions"]

```

```

fn shape(& self) -> Vec < Dimension >
{ self.external_dims_iter().collect() } fn reps(& self) -> Vec <
Representation < < Self :: Slot as IsAbstractSlot > :: R > >
{ self.external_reps_iter().collect() }
#[doc = " checks if externally, the two tensors are the same"] fn
same_external(& self, other : & Self) -> bool
{
    let set1 : HashSet < _ > = self.external_structure_iter().collect();
    let set2 : HashSet < _ > = other.external_structure_iter().collect();
    set1 == set2
}
#[doc =
" find the permutation of the external indices that would make the two tensors
the same. Applying the permutation to other should make it the same as self"]
fn find_permutation(& self, other : & Self) -> Result < Permutation >
{
    if self.order() != other.order()
    {
        return
        Err(eyre!
            ("Mismatched order: {} vs {}", self.order(), other.order()));
    } let other_structure = other.external_structure(); let self_structure
= self.external_structure(); let other_sort = Permutation ::
sort(& other_structure); let self_sort = Permutation ::
sort(& self_structure); if other_sort.apply_slice(& other_structure)
== self_sort.apply_slice(& self_structure)
{ Ok(other_sort.compose(& self_sort.inverse())) } else
{ Err(eyre! ("Mismatched structure")) }
} #[doc = " yields the strides of the tensor in column major order"] fn
strides_column_major(& self) -> Result < Vec < usize > >
{
    let mut strides : Vec < usize > = vec! [1; self.order()]; if
self.order() == 0 { return Ok(strides); } for i in 0 .. self.order() -
1
    {
        strides [i + 1] = strides [i] * usize ::
try_from(self.shape() [i]) ? ;
    } Ok(strides)
} #[doc = " yields the strides of the tensor in row major order"] fn
strides_row_major(& self) -> Result < Vec < usize > >
{
    let mut strides = vec! [1; self.order()]; if self.order() == 0
{ return Ok(strides); } for i in (0 .. self.order() - 1).rev()
    {
        strides [i] = strides [i + 1] * usize ::
try_from(self.shape() [i + 1]) ? ;
    } Ok(strides)
} #[doc = " By default, the strides are row major"] fn strides(& self) ->
Result < Vec < usize > > { self.strides_row_major() }
#[doc =
" Verifies that the list of indices provided are valid for the tensor"]
#[doc = ""] #[doc = " # Errors"] #[doc = ""]
#[doc =
" `Mismatched order` = if the length of the indices is different from the order
of the tensor,"]
#[doc = ""]

```



```

#[doc =
" `Index out of bounds` = if the index is out of bounds for the dimension of that
index"]
#[doc = ""] fn verify_indices < C : AsRef < [ConcreteIndex] > >
(& self, indices : C) -> Result < () >
{
    if indices.as_ref().len() != self.order()
    {
        return
        Err(eyre!
            ("Mismatched order: {} indices, vs order {}",
            indices.as_ref().len(), self.order()));
    } for (i, dim_len) in
self.external_structure_iter().map(| slot | slot.dim()).enumerate()
{
    if indices.as_ref() [i] >= usize :: try_from(dim_len) ?
    {
        return
        Err(eyre!
            ("Index {} out of bounds for dimension {} of size {}",
            indices.as_ref()[i], i, usize::try_from(dim_len)?));
    }
} Ok(())
} #[doc = " yields the flat index of the tensor given a list of indices"]
#[doc = ""] #[doc = " # Errors"] #[doc = ""]
#[doc = " Same as [`Self::verify_indices`]"] fn flat_index < C : AsRef <
[ConcreteIndex] > > (& self, indices : C) -> Result < FlatIndex >
{
    let strides = self.strides() ? ; self.verify_indices(& indices) ? ;
    let mut idx = 0; for (i, & index) in
indices.as_ref().iter().enumerate() { idx += index * strides [i]; }
    Ok(idx.into())
} #[doc = " yields the expanded index of the tensor given a flat index"]
#[doc = ""] #[doc = " # Errors"] #[doc = ""]
#[doc =
" `Index out of bounds` = if the flat index is out of bounds for the tensor"]
fn expanded_index(& self, flat_index : FlatIndex) -> Result <
ExpandedIndex >
{
    let mut indices = vec! []; let mut index : usize = flat_index.into();
    for & stride in & self.strides() ?
    { indices.push(index / stride); index %= stride; } if usize ::
from(flat_index) < self.size() ? { Ok(indices.into()) } else
    { Err(eyre! ("Index {flat_index} out of bounds")) }
} fn co_expanded_index(& self, flat_index : FlatIndex) -> Result <
DualConciousExpandedIndex >
{
    let mut indices = vec! []; for (r, i) in
self.external_reps_iter().zip(self.expanded_index(flat_index) ?
.iter())
{
    if r.rep.is_base() && r.rep.is_dual()
    { indices.push(DualConciousIndex :: SelfDual(* i)); } else if
r.rep.is_base() { indices.push(DualConciousIndex :: Up(* i)); }
    else { indices.push(DualConciousIndex :: Down(* i)); }
} Ok(indices.into())
}

```

```

    } #[doc = " yields an iterator over the indices of the tensor"] fn
    index_iter(& self) -> TensorStructureIndexIterator < '_, Self > where Self
    : Sized, { TensorStructureIndexIterator :: new(self) }
    #[doc = " if the tensor has no (external) indices, it is a scalar"] fn
    is_scalar(& self) -> bool { self.order() == 0 } fn
    is_fully_self_dual(& self) -> bool;
    #[doc =
    " yields the size of the tensor, i.e. the product of the dimensions. This is the
    length of the vector of the data in a dense tensor"]
    fn size(& self) -> Result < usize >
    {
        if self.order() == 0 { return Ok(1); } let mut size = 1; for dim in
        self.shape() { size *= usize :: try_from(dim) ? ; } Ok(size)
    }
}

```

Reference feature matrix: shadowing (item is available without an additional gate)

Supported usage

```
fn accepts_structure<S: spenso::structure::TensorStructure>(_structure: &S) {}
```

Slot

Kind: AssociatedType

```
type Slot : IsAbstractSlot + DualSlotTo < Dual = Self :: Slot > ;
```

Indexed

Kind: AssociatedType

```
type Indexed : TensorStructure < Indexed = Self :: Indexed, Slot = Self ::
Slot > ;
```

reindex

Kind: Method

```
fn reindex(self, indices : & [< Self :: Slot as IsAbstractSlot > :: Aind],) ->
Result < PermutedStructure < Self :: Indexed > , StructureError > ;
```

dual

Kind: Method

```
fn dual(self) -> Self;
```

string_rep

Kind: Method

```
fn string_rep(& self) -> String
{
    self.external_structure_iter().map(| s | s.to_string()).collect :: < Vec <
    _ > > ().join("")
}

```

external_structure_iter

Kind: Method

```
fn external_structure_iter(& self) -> impl Iterator < Item = Self :: Slot > ;
```

external_dims_iter

Kind: Method

```
fn external_dims_iter(& self) -> impl Iterator < Item = Dimension > ;
```

external_reps_iter

Kind: Method

```
fn external_reps_iter(& self,) -> impl Iterator < Item = Representation < < Self :: Slot as IsAbstractSlot > :: R > > ;
```

external_indices_iter

Kind: Method

```
fn external_indices_iter(& self) -> impl Iterator < Item = < Self :: Slot as IsAbstractSlot > :: Aind > ;
```

get_aind

Kind: Method

```
fn get_aind(& self, i : impl Into < SlotIndex >) -> Option < < Self :: Slot as IsAbstractSlot > :: Aind > ;
```

get_rep

Kind: Method

```
fn get_rep(& self, i : impl Into < SlotIndex > ,) -> Option < Representation < < Self :: Slot as IsAbstractSlot > :: R > > ;
```

get_dim

Kind: Method

```
fn get_dim(& self, i : impl Into < SlotIndex >) -> Option < Dimension > ;
```

get_slot

Kind: Method

```
fn get_slot(& self, i : impl Into < SlotIndex >) -> Option < Self :: Slot > ;
```

order

Kind: Method

```
fn order(& self) -> usize;
```

external_structure

Kind: Method

```
#[doc =  
" returns the list of slots that are the external indices of the tensor"] fn  
external_structure(& self) -> Vec < Self :: Slot >  
{ self.external_structure_iter().collect() }  
returns the list of slots that are the external indices of the tensor
```

to_shell

Kind: Method

```
fn to_shell(self) -> TensorShell < Self > where Self : Sized,  
{ TensorShell :: new(self) }
```

contains_matching

Kind: Method

```
fn contains_matching(& self, slot : & Self :: Slot) -> bool
{ self.external_structure_iter().any(| s | s.matches(slot)) }
```

external_reps

Kind: Method

```
fn external_reps(& self) -> Vec < Representation < < Self :: Slot as
IsAbstractSlot > :: R > > { self.external_reps_iter().collect() }
```

external_indices

Kind: Method

```
fn external_indices(& self) -> Vec < < Self :: Slot as IsAbstractSlot > ::
Aind > { self.external_indices_iter().collect() }
```

same_content

Kind: Method

```
#[doc =
" checks if the tensor has the same exact structure as another tensor"] fn
same_content(& self, other : & Self) -> bool { self.same_external(other) }
checks if the tensor has the same exact structure as another tensor
```

match_index

Kind: Method

```
#[doc =
" Given two [`TensorStructure`]s, returns the index of the first matching slot in
each external index list, along with a boolean indicating if there is a single
match"]
fn match_index(& self, other : & Self) -> Option < (bool, usize, usize) >
{
    let posmap =
self.external_structure_iter().enumerate().map(| (i, slot) |
(slot, i)).collect :: < AHashMap < _, _ > > (); let mut first_pair :
Option < (usize, usize) > = None; for (j, slot) in
other.external_structure_iter().enumerate()
{
    if let Some(& i) = posmap.get(& slot.dual())
    {
        if let Some((i, j)) = first_pair { return Some((false, i, j)); }
        first_pair = Some((i, j));
    }
} first_pair.map(| (i, j) | (true, i, j))
}
```

Given two [`TensorStructure`]s, returns the index of the first matching slot in each external index list, along with a boolean indicating if there is a single match

match_indices

Kind: Method

```
#[doc =
" Given two [`TensorStructure`]s, returns the index of the first matching slot in
each external index list"]
```

```

fn match_indices(& self, other : & Self) -> Option <
(Permutation, Vec < bool > , Vec < bool >) >
{
    let mut self_matches = vec! [false; self.order()]; let mut perm = Vec ::
new(); let mut other_matches = vec! [false; other.order()]; let posmap =
self.external_structure_iter().enumerate().map(| (i, slot) |
(slot, i)).collect :: < AHashMap < _, _ > > (); for (j, slot_other) in
other.external_structure_iter().enumerate()
{
    if let Some(& i) = posmap.get(& slot_other.dual())
    { self_matches [i] = true; other_matches [j] = true; perm.push(i); }
} if perm.is_empty() { None } else
{
    let p : Permutation = Permutation :: sort(& perm);
    Some((p, self_matches, other_matches))
}
}

```

Given two [`TensorStructure`]'s, returns the index of the first matching slot in each external index list

traces

Kind: Method

```

#[doc = " Identify the repeated slots in the external index list"] fn
traces(& self) -> Vec < [usize; 2] >
{
    let mut positions : HashMap < < Self as TensorStructure > :: Slot, Vec <
usize > > = HashMap :: new(); for (index, key) in
self.external_structure_iter().enumerate()
{
    if let Some(v) = positions.get_mut(& key.dual()) { v.push(index); }
    else { positions.insert(key, vec! [index]); }
}
positions.into_values().filter_map(| indices |
{
    if indices.len() == 2 { Some([indices [0], indices [1]]) } else
    { None }
}).collect()
}

```

Identify the repeated slots in the external index list

shape

Kind: Method

```

#[doc =
" yields the (outwards facing) shape of the tensor as a list of dimensions"]
fn shape(& self) -> Vec < Dimension > { self.external_dims_iter().collect() }
yields the (outwards facing) shape of the tensor as a list of dimensions

```

reps

Kind: Method

```

fn reps(& self) -> Vec < Representation < < Self :: Slot as IsAbstractSlot >
:: R > > { self.external_reps_iter().collect() }

```

same_external

Kind: Method

```
#[doc = " checks if externally, the two tensors are the same"] fn
same_external(& self, other : & Self) -> bool
{
    let set1 : HashSet < _ > = self.external_structure_iter().collect(); let
    set2 : HashSet < _ > = other.external_structure_iter().collect(); set1 ==
    set2
}
```

checks if externally, the two tensors are the same

find_permutation

Kind: Method

```
#[doc =
" find the permutation of the external indices that would make the two tensors the
same. Applying the permutation to other should make it the same as self"]
fn find_permutation(& self, other : & Self) -> Result < Permutation >
{
    if self.order() != other.order()
    {
        return
        Err(eyre!
            ("Mismatched order: {} vs {}", self.order(), other.order()));
    } let other_structure = other.external_structure(); let self_structure =
    self.external_structure(); let other_sort = Permutation ::
    sort(& other_structure); let self_sort = Permutation ::
    sort(& self_structure); if other_sort.apply_slice(& other_structure) ==
    self_sort.apply_slice(& self_structure)
    { Ok(other_sort.compose(& self_sort.inverse())) } else
    { Err(eyre! ("Mismatched structure")) }
}
```

find the permutation of the external indices that would make the two tensors the same. Applying the permutation to other should make it the same as self

strides_column_major

Kind: Method

```
#[doc = " yields the strides of the tensor in column major order"] fn
strides_column_major(& self) -> Result < Vec < usize > >
{
    let mut strides : Vec < usize > = vec! [1; self.order()]; if self.order()
    == 0 { return Ok(strides); } for i in 0 .. self.order() - 1
    {
        strides [i + 1] = strides [i] * usize :: try_from(self.shape() [i]) ?
        ;
    } Ok(strides)
}
```

yields the strides of the tensor in column major order

strides_row_major

Kind: Method

```
#[doc = " yields the strides of the tensor in row major order"] fn
strides_row_major(& self) -> Result < Vec < usize > >
```

```

{
    let mut strides = vec! [1; self.order()]; if self.order() == 0
    { return Ok(strides); } for i in (0 .. self.order() - 1).rev()
    {
        strides [i] = strides [i + 1] * usize ::
            try_from(self.shape() [i + 1]) ? ;
    } Ok(strides)
}

```

yields the strides of the tensor in row major order

strides

Kind: Method

```

#[doc = " By default, the strides are row major"] fn strides(& self) -> Result
< Vec < usize > > { self.strides_row_major() }

```

By default, the strides are row major

verify_indices

Kind: Method

```

#[doc =
" Verifies that the list of indices provided are valid for the tensor"
#[doc = "" ] #[doc = " # Errors" ] #[doc = "" ]
#[doc =
" `Mismatched order` = if the length of the indices is different from the order of
the tensor," ]
#[doc = "" ]
#[doc =
" `Index out of bounds` = if the index is out of bounds for the dimension of that
index" ]
#[doc = "" ] fn verify_indices < C : AsRef < [ConcreteIndex] > >
(& self, indices : C) -> Result < () >
{
    if indices.as_ref().len() != self.order()
    {
        return
        Err(eyre!
            ("Mismatched order: {} indices, vs order {}", indices.as_ref().len(),
            self.order()));
    } for (i, dim_len) in
self.external_structure_iter().map(| slot | slot.dim()).enumerate()
{
    if indices.as_ref() [i] >= usize :: try_from(dim_len) ?
    {
        return
        Err(eyre!
            ("Index {} out of bounds for dimension {} of size {}",
            indices.as_ref()[i], i, usize::try_from(dim_len)?));
    }
}
} Ok(())
}

```

Verifies that the list of indices provided are valid for the tensor

Errors

`Mismatched order` = if the length of the indices is different from the order of the

tensor,

`Index out of bounds` = if the index is out of bounds for the dimension of that index

flat_index

Kind: Method

```
#[doc = " yields the flat index of the tensor given a list of indices"]
#[doc = ""] #[doc = " # Errors"] #[doc = ""]
#[doc = " Same as [`Self::verify_indices`]"] fn flat_index < C : AsRef <
[ConcreteIndex] > > (& self, indices : C) -> Result < FlatIndex >
{
    let strides = self.strides() ? ; self.verify_indices(& indices) ? ; let
    mut idx = 0; for (i, & index) in indices.as_ref().iter().enumerate()
    { idx += index * strides [i]; } Ok(idx.into())
}
```

yields the flat index of the tensor given a list of indices

Errors

Same as [`Self::verify\_indices`]

expanded_index

Kind: Method

```
#[doc = " yields the expanded index of the tensor given a flat index"]
#[doc = ""] #[doc = " # Errors"] #[doc = ""]
#[doc =
" `Index out of bounds` = if the flat index is out of bounds for the tensor"]
fn expanded_index(& self, flat_index : FlatIndex) -> Result < ExpandedIndex >
{
    let mut indices = vec! []; let mut index : usize = flat_index.into(); for
    & stride in & self.strides() ?
    { indices.push(index / stride); index %= stride; } if usize ::
    from(flat_index) < self.size() ? { Ok(indices.into()) } else
    { Err(eyre! ("Index {flat_index} out of bounds")) }
}
```

yields the expanded index of the tensor given a flat index

Errors

`Index out of bounds` = if the flat index is out of bounds for the tensor

co_expanded_index

Kind: Method

```
fn co_expanded_index(& self, flat_index : FlatIndex) -> Result <
DualConciousExpandedIndex >
{
    let mut indices = vec! []; for (r, i) in
    self.external_reps_iter().zip(self.expanded_index(flat_index) ? .iter())
    {
        if r.rep.is_base() && r.rep.is_dual()
        { indices.push(DualConciousIndex :: SelfDual(* i)); } else if
        r.rep.is_base() { indices.push(DualConciousIndex :: Up(* i)); } else
        { indices.push(DualConciousIndex :: Down(* i)); }
    }
```



```
    } Ok(indices.into())
}
```

index_iter

Kind: Method

```
#[doc = " yields an iterator over the indices of the tensor"] fn
index_iter(& self) -> TensorStructureIndexIterator < '_, Self > where Self :
Sized, { TensorStructureIndexIterator :: new(self) }

yields an iterator over the indices of the tensor
```

is_scalar

Kind: Method

```
#[doc = " if the tensor has no (external) indices, it is a scalar"] fn
is_scalar(& self) -> bool { self.order() == 0 }

if the tensor has no (external) indices, it is a scalar
```

is_fully_self_dual

Kind: Method

```
fn is_fully_self_dual(& self) -> bool;
```

size

Kind: Method

```
#[doc =
" yields the size of the tensor, i.e. the product of the dimensions. This is the
length of the vector of the data in a dense tensor"]
fn size(& self) -> Result < usize >
{
    if self.order() == 0 { return Ok(1); } let mut size = 1; for dim in
    self.shape() { size *= usize :: try_from(dim) ? ; } Ok(size)
}
```

yields the size of the tensor, i.e. the product of the dimensions. This is the length of the vector of the data in a dense tensor

[Exhaustive reference](#) · [Source](#)

Dense tensor

Stores every tensor component in structure-defined flat order.

Stores every tensor component in structure-defined flat order.

```
#[derive(Debug, Clone, PartialEq, Deserialize, Serialize, Hash, Eq, Encode,
Decode)] pub struct DenseTensor < T, S = OrderedStructure >
{ pub data : Vec < T > , pub structure : S, }
```

Reference feature matrix: shadowing (item is available without an additional gate)

Supported usage

```
use spenso::tensors::data::DenseTensor;
```

data

Kind: Field

Vec < T >

structure

Kind: Field

S

[Exhaustive reference](#) · [Source](#)

Sparse tensor

Stores only explicit tensor components keyed by flat index.

Sparse data tensor, generic on storage type `T`, and structure type `I`.

Stores data in a hashmap of usize, using ahash's hashmap.

The usize key is the flattened index of the corresponding position in the dense tensor

```
#[doc =  
" Sparse data tensor, generic on storage type `T`, and structure type `I`."]  
#[doc = ""]  
#[doc = " Stores data in a hashmap of usize, using ahash's hashmap."]  
#[doc =  
" The usize key is the flattened index of the corresponding position in the dense  
tensor"]  
#[derive(Debug, Clone, Serialize, Deserialize, PartialEq, Eq, Encode, Decode)]  
pub struct SparseTensor < T, I = OrderedStructure >  
{  
    pub elements : std :: collections :: HashMap < FlatIndex, T > , pub zero :  
        T, pub structure : I,  
}
```

Reference feature matrix: shadowing (item is available without an additional gate)

Supported usage

```
use spenso::tensors::data::SparseTensor;
```

elements

Kind: Field

std :: collections :: HashMap < FlatIndex, T >

zero

Kind: Field

T

structure

Kind: Field

I

[Exhaustive reference](#) · [Source](#)

Pairwise contraction

Defines pairwise tensor contraction with an explicit settings type and output.

Defines pairwise tensor contraction with an explicit settings type and output.

```
pub trait Contract < T = Self, Settings = () >  
{  
    type LCM; fn contract(& self, other : & T) -> Result < Self :: LCM,
```

```

    ContractionError > ;
}

```

Reference feature matrix: shadowing (item is available without an additional gate)

Supported usage

```
fn can_contract<T: spenso::contraction::Contract>(_tensor: &T) {}
```

LCM

Kind: AssociatedType

```
type LCM;
```

contract

Kind: Method

```
fn contract(& self, other : & T) -> Result < Self :: LCM, ContractionError > ;
```

Exhaustive reference · Source

Tensor network

Binds symbolic tensor expressions to libraries and an executable contraction graph.

Binds symbolic tensor expressions to libraries and an executable contraction graph.

```
#[derive(Debug, Clone, Serialize, Deserialize, bincode_trait_derive::Encode,
bincode_trait_derive::Decode, PartialEq, Eq,)]
#[cfg_attr(feature = "shadowing",
trait_decode(trait = symbolica::state::HasStateMap),)] pub struct Network < S,
LibKey, FunKey, Aind = AbstractIndex >
{
    pub graph : NetworkGraph < LibKey, FunKey, Aind > , pub store : S, pub
state : NetworkState,
}
```

Reference feature matrix: shadowing (item is available without an additional gate)

Supported usage

```
use spenso::network::Network;
```

graph

Kind: Field

```
NetworkGraph < LibKey, FunKey, Aind >
```

store

Kind: Field

```
S
```

state

Kind: Field

```
NetworkState
```

Exhaustive reference · Source

Parametric tensor

Stores tensor components as parameter-dependent symbolic coefficients.

Stores tensor components as parameter-dependent symbolic coefficients.

```
#[derive(Clone, Debug, PartialEq, Eq, Hash, bincode_trait_derive::Encode,
bincode_trait_derive::Decode,)]
#[trait_decode(trait = symbolica::state::HasStateMap)] pub struct ParamTensor
< S = OrderedStructure >
{ pub tensor : DataTensor < Atom, S > , pub param_type : ParamOrComposite, }
```

Requires features: shadowing

Supported usage

```
use spenso::tensors::parametric::ParamTensor;
```

tensor

Kind: Field

DataTensor < Atom, S >

param_type

Kind: Field

ParamOrComposite

[Exhaustive reference](#) · [Source](#)

spenso-macros

Spenso's exported procedural macros. Expansion details remain in the exhaustive Rustdoc sidecar.

SimpleRepresentation derive

Derives representation naming and duality boilerplate from the representation helper attribute.

Derives representation naming and duality boilerplate from the representation helper attribute.

```
#[derive(SimpleRepresentation)]
fn derive_simple_representation(input : TokenStream) -> TokenStream
```

Reference feature matrix: shadowing (item is available without an additional gate)

Supported usage

```
#[derive(Debug, Clone, Copy, PartialEq, Eq, Hash, PartialOrd, Ord, Default,
spenso_macros::SimpleRepresentation)] #[representation(name = "flavor", self_dual)]
struct Flavor;
```

[Exhaustive reference](#) · [Source](#)

spenso-hep-lib

Spenso's high-energy-physics companion library. These entries *complement* the tensor concepts owned by the main Spenso documentation.

HEP tensor library

Constructs the standard high-energy-physics tensor library for a chosen scalar data type.

Constructs the standard high-energy-physics tensor library for a chosen scalar data type.

```
fn hep_lib < Aind : AbsInd, T : TensorLibraryData + Clone + Default >
(one : T, zero : T,) -> TensorLibrary < MixedTensor < T, ExplicitKey < Aind >
> , Aind >
```

Parameter	Type
one	T
zero	T

Returns: TensorLibrary < MixedTensor < T, ExplicitKey < Aind > > , Aind >

Supported usage

```
use spenso_hep_lib::hep_lib;
```

Exhaustive reference · Source

Dirac gamma data

Creates sparse Dirac-basis gamma-matrix data for the supplied tensor structure.

Creates sparse Dirac-basis gamma-matrix data for the supplied tensor structure.

```
fn gamma_data_dirac < T, N > (structure : N, one : T, zero : T) ->
SparseTensor < Complex < T > , N > where T : Clone + Neg < Output = T > , N :
TensorStructure,
```

Parameter	Type
structure	N
one	T
zero	T

Returns: SparseTensor < Complex < T > , N >

Supported usage

```
use spenso_hep_lib::gamma_data_dirac;
```

Exhaustive reference · Source

SU(3) generator data

Creates the standard sparse SU(3) fundamental-generator tensor.

Fundamental SU(3) generators in the normalization $\text{Tr}(T^a T^b) = 1/2 \delta^{ab}$.

The index order follows `CS.t_strct`: adjoint, fundamental, anti-fundamental.

```
fn su3_generator_data < N > (structure : N) -> SparseTensor < Complex < f64 >
, N > where N : TensorStructure,
```

Parameter	Type
structure	N

Returns: SparseTensor < Complex < f64 > , N >

Supported usage

```
use spenso_hep_lib::su3_generator_data;
```

Exhaustive reference · Source

spynso3

Exports registered by spynso3.

Representation

A representation in the sense of group representation theory for tensor indices.

A representation in the sense of group representation theory for tensor indices.

Representations define the transformation properties of tensor indices under group operations.

They specify the dimension and duality structure, determining which indices can contract.

Key concepts:

- **Self-dual**: Indices can contract with other indices of the same representation
- **Dualizable**: Indices can only contract with their dual representation
- **Dimension**: Size of the representation space

Predefined representations are available as class methods:

- `Representation.euc(d)`: Euclidean space (self-dual)
- `Representation.mink(d)`: Minkowski space (self-dual)
- `Representation.bis(d)`: Bispinor (self-dual)
- `Representation.cof(d)`: Color fundamental (dualizable)
- `Representation.coad(d)`: Color adjoint (self-dual)
- `Representation.cos(d)`: Color sextet (dualizable)

Examples:

```
```python
from symbolica.community.spenso import Representation

Standard representations
euclidean = Representation.euc(4) # 4D Euclidean
lorentz = Representation.mink(4) # 4D Minkowski
color = Representation.cof(3) # SU(3) fundamental
adjoint = Representation.coad(8) # SU(3) adjoint

Custom representation
custom = Representation("MyRep", 5, is_self_dual=True)

Create slots with indices
mu_slot = euclidean('mu') # Euclidean index μ
a_slot = color('a') # Color index a

Generate metric tensors
metric = euclidean.g('mu', 'nu') # $g_{\mu\nu}$
```

class Representation:
```

Requires features: python_stubgen

Inspect the registered export

```
from symbolica.community.spenso import Representation
help(Representation)
```

`__call__`

Kind: Method

```
def __call__(self, aind: builtins.int | Expression | str) -> Slot:
```

Create a slot from this representation, by specifying an index.

Parameters

a_{ind} : int, str, or Symbol

The index specification

Returns

Slot

A new Slot object with the specified index

Examples

```
>>> from symbolica.community.spenso import Representation
>>> import symbolica as sp
>>> rep = Representation.euc(3)
>>> slot1 = rep('mu')
>>> slot2 = rep(1)
>>> slot3 = rep(sp.S('nu'))
```

a_{ind}

Kind: Parameter

a_{ind}: builtins.int | Expression | str

The index specification

__call__

Kind: Method

def **__call__**(self, a_{ind}: Expression) -> Expression | Slot:

Create a slot or symbolic expression from this representation.

Parameters

a_{ind} : Expression

The index specification (Expression creates symbolic representation)

Returns

Expression

A symbolic expression representing this representation

Examples

```
>>> from symbolica.community.spenso import Representation
>>> import symbolica as sp
>>> rep = Representation.euc(3)
>>> expr = rep(sp.E("cos(x)"))
```

a_{ind}

Kind: Parameter

a_{ind}: Expression

The index specification (Expression creates symbolic representation)

__new__

Kind: AssociatedFunction

```
def __new__(cls, name: builtins.str, dimension: builtins.int | Expression | str,
is_self_dual: builtins.bool = ...) -> Representation:
```

Create and register a new representation with specified properties.

Parameters:

- name: String name for the representation
- dimension: Size of the representation (int or symbolic)
- is_self_dual: If True, creates self-dual representation; if False, creates dualizable pair

Examples:

```
```python
from symbolica.community.spenso import Representation
import symbolica as sp

Self-dual representation (indices contract with themselves)
euclidean = Representation("Euclidean", 4, is_self_dual=True)

Dualizable representation (needs dual partner for contraction)
vector_up = Representation("VectorUp", 4, is_self_dual=False)

Symbolic dimension
n = sp.S('n')
general = Representation("General", n, is_self_dual=True)
```
```

name

Kind: Parameter

name: builtins.str

dimension

Kind: Parameter

dimension: builtins.int | Expression | str

is_self_dual

Kind: Parameter

is_self_dual: builtins.bool = ...

Default: ...

dual

Kind: Method

```
def dual(self) -> Representation:
```

g

Kind: Method

```
def g(self, i: builtins.int | Expression | str, j: builtins.int | Expression | str) -
> TensorIndices:
```

Create a metric tensor for this representation.


```

# Parameters:
- i: First index
- j: Second index

# Examples:
```python
rep = Representation.mink(4)
metric = rep.g('mu', 'nu') # Minkowski metric $g_{\mu\nu}$
```

```

i

Kind: Parameter

i: builtins.int | Expression | str

j

Kind: Parameter

j: builtins.int | Expression | str

flat

Kind: Method

```

def flat(self, i: builtins.int | Expression | str, j: builtins.int | Expression | str) -> TensorIndices:

```

Create a musical isomorphism tensor for this representation.

```

# Parameters:
- i: First index
- j: Second index

```

```

# Examples:
```python
rep = Representation.mink(4)
flat = rep.flat('mu', 'nu') # Flat isomorphism $b_{\mu\nu}$
```

```

i

Kind: Parameter

i: builtins.int | Expression | str

j

Kind: Parameter

j: builtins.int | Expression | str

id

Kind: Method

```

def id(self, i: builtins.int | Expression | str, j: builtins.int | Expression | str) -> TensorIndices:

```

Create an identity tensor for this representation.

```

# Parameters:
- i: First index
- j: Second index

```

```
# Examples:
```python
rep = Representation.cof(3)
identity = rep.id('a', 'b') # Color identity δ_{ab}
```
```

i

Kind: Parameter

i: builtins.int | Expression | str

j

Kind: Parameter

j: builtins.int | Expression | str

__repr__

Kind: Method

def __repr__(self) -> builtins.str:

__str__

Kind: Method

def __str__(self) -> builtins.str:

to_expression

Kind: Method

def to_expression(self) -> Expression:

Convert the representation to a symbolic expression.

bis

Kind: AssociatedFunction

def bis(dimension: builtins.int | Expression | str) -> Representation:

Create a bispinor representation.

Parameters:

- dimension: The dimension of the bispinor space

dimension

Kind: Parameter

dimension: builtins.int | Expression | str

euc

Kind: AssociatedFunction

def euc(dimension: builtins.int | Expression | str) -> Representation:

Create a Euclidean space representation.

Parameters:

- dimension: The dimension of the Euclidean space

dimension

Kind: Parameter

dimension: builtins.int | Expression | str

mink

Kind: AssociatedFunction

def mink(dimension: builtins.int | Expression | str) -> Representation:

Create a Minkowski space representation.

Parameters:

- dimension: The dimension of the Minkowski space

dimension

Kind: Parameter

dimension: builtins.int | Expression | str

cof

Kind: AssociatedFunction

def cof(dimension: builtins.int | Expression | str) -> Representation:

Create a color fundamental representation.

Parameters:

- dimension: The dimension of the color group (e.g., 3 for SU(3))

dimension

Kind: Parameter

dimension: builtins.int | Expression | str

coad

Kind: AssociatedFunction

def coad(dimension: builtins.int | Expression | str) -> Representation:

Create a color adjoint representation.

Parameters:

- dimension: The dimension of the adjoint representation (e.g., 8 for SU(3))

dimension

Kind: Parameter

dimension: builtins.int | Expression | str

cos

Kind: AssociatedFunction

def cos(dimension: builtins.int | Expression | str) -> Representation:

Create a color sextet representation.

Parameters:

- dimension: The dimension of the sextet representation (e.g., 6 for SU(3))

dimension

Kind: Parameter

dimension: builtins.int | Expression | str

Exhaustive reference · Source

Slot

A tensor index slot combining a representation with an abstract index.

A tensor index slot combining a representation with an abstract index.

Slots are the building blocks for tensor structures, pairing a representation (which defines transformation properties) with an abstract index identifier. Slots with matching representations and indices can be contracted.

Examples:

```
```python
from symbolica.community.spenso import Slot, Representation
import symbolica as sp
```

# Create representation and slots

```
rep = Representation.euc(3)
slot1 = rep('mu') # Slot with string index
slot2 = rep(1) # Slot with integer index
slot3 = rep(sp.S('nu')) # Slot with symbolic index
```

# Create custom slot

```
custom_slot = Slot("MyRep", 4, 'alpha', dual=False)
```

# Use in tensor structures

```
from symbolica.community.spenso import TensorIndices
tensor_structure = TensorIndices(slot1, slot2)
```
```

```
class Slot:
```

Requires features: python_stubgen

Inspect the registered export

```
from symbolica.community.spenso import Slot
help(Slot)
```

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

__str__

Kind: Method

```
def __str__(self) -> builtins.str:
```

dual

Kind: Method

```
def dual(self) -> Slot:
```

__new__

Kind: AssociatedFunction

```
def __new__(cls, name: builtins.str, dimension: builtins.int, aind: builtins.int |
Expression | str, dual: builtins.bool = ...) -> Slot:
```

Create a new slot with a custom representation and index.

Parameters:

- name: String name for the representation
- dimension: Size of the representation space
- aind: The abstract index (int, str, or Symbol)
- dual: If True, creates dualizable representation; if False, self-dual

Examples:

```
```python
from symbolica.community.spenso import Slot
import symbolica as sp

Self-dual slot
euclidean_slot = Slot("Euclidean", 4, 'mu', dual=False)

Dualizable slot
vector_slot = Slot("Vector", 4, 'nu', dual=True)

With symbolic index
sym_index = sp.S('alpha')
symbolic_slot = Slot("Custom", 3, sym_index, dual=False)
```
```

name

Kind: Parameter

name: builtins.str

dimension

Kind: Parameter

dimension: builtins.int

aind

Kind: Parameter

aind: builtins.int | Expression | str

dual

Kind: Parameter

dual: builtins.bool = ...

Default: ...

to_expression

Kind: Method

def to_expression(self) -> Expression:

Convert the slot to a symbolic expression.

Examples:

```
```python
rep = Representation.euc(3)
slot = rep('mu')
expr = slot.to_expression() # Symbolic representation of the slot
```
```

Exhaustive reference · Source

Tensor

A tensor class that can be either dense or sparse with flexible data types.

A tensor class that can be either dense or sparse with flexible data types.

The tensor can store data as floats, complex numbers, or symbolic expressions.

Tensors have an associated structure that defines their shape and index properties.

Examples

```
-----
>>> from symbolica.community.spenso import Tensor, TensorIndices, Representation
>>> structure = TensorIndices(Representation.euc(4)(1))
>>> data = [1.0, 2.0, 3.0, 4.0]
>>> tensor = Tensor.dense(structure, data)
>>> sparse_tensor = Tensor.sparse(structure, float)
```

```
class Tensor:
```

Requires features: python_stubgen

Inspect the registered export

```
from symbolica.community.spenso import Tensor
help(Tensor)
```

structure

Kind: Method

```
def structure(self) -> TensorIndices:
```

sparse

Kind: AssociatedFunction

```
def sparse(structure: TensorIndices | builtins.list[Slot], type_info: type) ->
Tensor:
```

Create a new sparse empty tensor with the given structure and data type.

Parameters

```
-----
structure : TensorIndices or list of Slots
The tensor structure defining shape and index properties
type_info : type
The data type - either `float` or `Expression` class
```

Returns

```
-----
Tensor
A new sparse tensor with all elements initially zero
```

Examples

```
-----
>>> from symbolica.community.spenso import Tensor, TensorIndices, Representation as R
>>> structure = TensorIndices(R.euc(3)(1), R.euc(3)(2))
>>> sparse_float = Tensor.sparse(structure, float)
>>> sparse_sym = Tensor.sparse(structure, symbolica.Expression)
```

structure

Kind: Parameter

structure: TensorIndices | builtins.list[Slot]

The tensor structure defining shape and index properties

type_info

Kind: Parameter

type_info: type

The data type - either `float` or `Expression` class

dense

Kind: AssociatedFunction

```
def dense(structure: TensorIndices | builtins.list[Slot], data:
typing.Sequence[Expression] | typing.Sequence[builtins.float] |
typing.Sequence[builtins.complex]) -> Tensor:
```

Create a new dense tensor with the given structure and data.

Parameters

structure : TensorIndices or list of Slots

The tensor structure defining shape and index properties

data : list of float, complex, or Expression

The tensor data in row-major order

Returns

Tensor

A new dense tensor with the specified data

Examples

```
>>> from symbolica import S
>>> from symbolica.community.spenso import Tensor, TensorIndices, Representation as R
>>> structure = TensorIndices(R.euc(2)(1), R.euc(2)(2))
>>> data = [1.0, 2.0, 3.0, 4.0]
>>> tensor = Tensor.dense(structure, data)
>>> x, y = S("x", "y")
>>> sym_data = [x, y, x * y, x + y]
>>> sym_tensor = Tensor.dense(structure, sym_data)
```

structure

Kind: Parameter

structure: TensorIndices | builtins.list[Slot]

The tensor structure defining shape and index properties

data

Kind: Parameter

data: typing.Sequence[Expression] | typing.Sequence[builtins.float] |
typing.Sequence[builtins.complex]

The tensor data in row-major order

one

Kind: AssociatedFunction

```
def one() -> Tensor:
```

Create a scalar tensor with value 1.0.

Returns

Tensor

A scalar tensor containing the value 1.0

Examples

```
>>> from symbolica.community.spenso import Tensor
>>> one = Tensor.one()
```

zero

Kind: AssociatedFunction

```
def zero() -> Tensor:
```

Create a scalar tensor with value 0.0.

Returns

Tensor

A scalar tensor containing the value 0.0

Examples

```
>>> from symbolica.community.spenso import Tensor
>>> zero = Tensor.zero()
```

to_dense

Kind: Method

```
def to_dense(self) -> None:
```

Convert this tensor to dense storage format.

Convert this tensor to dense storage format.

Converts sparse tensors to dense format in-place. Dense tensors are unchanged.

This allocates memory for all tensor elements.

Examples

```
>>> from symbolica.community.spenso import Tensor, TensorIndices, Representation as R
>>> structure = TensorIndices(R.euc(4)(2))
>>> tensor = Tensor.sparse(structure, float)
>>> tensor[0] = 1.0
>>> tensor.to_dense()
```

to_sparse

Kind: Method

```
def to_sparse(self) -> None:
```


Convert this tensor to sparse storage format.

Convert this tensor to sparse storage format.

Converts dense tensors to sparse format in-place, only storing non-zero elements.
This can save memory for tensors with many zero elements.

Examples

```
>>> from symbolica.community.spenso import Tensor, TensorIndices, Representation as R
>>> structure = TensorIndices(R.euc(2)(2), R.euc(2)(1))
>>> data = [1.0, 0.0, 0.0, 2.0]
>>> tensor = Tensor.dense(structure, data)
>>> tensor.to_sparse()
```

`__repr__`

Kind: Method

```
def __repr__(self) -> builtins.str:
```

`__str__`

Kind: Method

```
def __str__(self) -> builtins.str:
```

`__len__`

Kind: Method

```
def __len__(self) -> builtins.int:
```

`__getitem__`

Kind: Method

```
def __getitem__(self, item: builtins.slice | builtins.int |
builtins.list[builtins.int]) -> typing.Any:
```

`item`

Kind: Parameter

```
item: builtins.slice | builtins.int | builtins.list[builtins.int]
```

`__getitem__`

Kind: Method

```
def __getitem__(self, item: builtins.slice) -> builtins.list[Expression |
builtins.complex | float]:
```

Get tensor elements at the specified range of indices.

Parameters

item : slice

Slice object defining the range of indices

Returns

list of float, complex, or Expression

The tensor elements at the specified range

item

Kind: Parameter

item: builtins.slice

Slice object defining the range of indices

__getitem__

Kind: Method

```
def __getitem__(self, item: typing.Sequence[builtins.int] | builtins.int) ->
Expression | builtins.complex | float:
```

Get tensor element at the specified index or indices.

Parameters

item : int or list of int

Index specification (int for flat index, list of int for coordinates)

Returns

float, complex, or Expression

The tensor element at the specified index

item

Kind: Parameter

item: typing.Sequence[builtins.int] | builtins.int

Index specification (int for flat index, list of int for coordinates)

__setitem__

Kind: Method

```
def __setitem__(self, item: typing.Any, value: typing.Any) -> None:
```

Set tensor element(s) at the specified index or indices.

Parameters

item : int or list of int

Index specification (int for flat index, list of int for coordinates)

value : float, complex, or Expression

The value to set

Examples

```
>>> from symbolica.community.spenso import Tensor, TensorIndices, Representation as R
>>> structure = TensorIndices(R.euc(2)(2), R.euc(2)(1))
>>> tensor = Tensor.sparse(structure, float)
>>> tensor[0] = 4.0
>>> tensor[1, 1] = 1.0
```

item

Kind: Parameter

item: typing.Any

Index specification (int for flat index, list of int for coordinates)

value

Kind: Parameter

value: typing.Any

The value to set

__setitem__

Kind: Method

```
def __setitem__(self, item: builtins.int | typing.Sequence[builtins.int], value: Expression | builtins.complex | float) -> None:
```

Set tensor element at the specified index.

Parameters

item : int or list of int

Index specification (int for flat index, list of int for coordinates)

value : float, complex, or Expression

The value to set

Examples

```
>>> from symbolica.community.spenso import Tensor, TensorIndices, Representation
>>> rep = Representation.euc(2)
>>> structure = TensorIndices(rep(1), rep(2))
>>> tensor = Tensor.sparse(structure, float)
>>> tensor[0] = 1.0
>>> tensor[1, 1] = 2.0
```

item

Kind: Parameter

item: builtins.int | typing.Sequence[builtins.int]

Index specification (int for flat index, list of int for coordinates)

value

Kind: Parameter

value: Expression | builtins.complex | float

The value to set

evaluator

Kind: Method

```
def evaluator(self, constants: typing.Mapping[Expression, Expression], funs: typing.Mapping[tuple[Expression, builtins.str, typing.Sequence[Expression]], Expression], params: typing.Sequence[Expression], iterations: builtins.int = ..., n_cores: builtins.int = ..., verbose: builtins.bool = ...) -> TensorEvaluator:
```

Create an optimized evaluator for symbolic tensor expressions.

Create an optimized evaluator for symbolic tensor expressions.

Compiles the symbolic expressions in this tensor into an optimized evaluation tree that can efficiently compute numerical values for different parameter inputs.

Parameters

```

-----
constants : dict
Dict mapping symbolic expressions to their constant numerical values
funs : dict
Dict mapping function signatures to their symbolic definitions
params : list of Expression
List of symbolic parameters that will be varied during evaluation
iterations : int, optional
Number of optimization iterations for Horner scheme (default: 100)
n_cores : int, optional
Number of CPU cores to use for optimization (default: 4)
verbose : bool, optional
Whether to print optimization progress (default: False)

```

Returns

```

-----
TensorEvaluator
An optimized evaluator for efficient numerical evaluation

```

Examples

```

-----
>>> from symbolica import S
>>> from symbolica.community.spenso import Tensor, TensorIndices, Representation as R
>>> x, y = S("x", "y")
>>> structure = TensorIndices(R.euc(2)(1))
>>> tensor = Tensor.dense(structure, [x * y, x + y])
>>> evaluator = tensor.evaluator(constants={}, funs={}, params=[x, y], iterations=50)
>>> results = evaluator.evaluate_complex([[1.0, 2.0], [3.0, 4.0]])

```

constants

Kind: Parameter

constants: typing.Mapping[Expression, Expression]

Dict mapping symbolic expressions to their constant numerical values

funs

Kind: Parameter

funs: typing.Mapping[tuple[Expression, builtins.str, typing.Sequence[Expression]], Expression]

Dict mapping function signatures to their symbolic definitions

params

Kind: Parameter

params: typing.Sequence[Expression]

List of symbolic parameters that will be varied during evaluation

iterations

Kind: Parameter

iterations: builtins.int = ...

Default: ...

Number of optimization iterations for Horner scheme (default: 100)

n_cores

Kind: Parameter

n_cores: builtins.int = ...

Default: ...

Number of CPU cores to use for optimization (default: 4)

verbose

Kind: Parameter

verbose: builtins.bool = ...

Default: ...

Whether to print optimization progress (default: False)

scalar

Kind: Method

def scalar(self) -> Expression:

Extract the scalar value from a rank-0 (scalar) tensor.

Returns

Expression

The scalar expression contained in this tensor

Raises

RuntimeError

If the tensor is not a scalar

Examples

```
>>> from symbolica.community.spenso import Tensor
```

```
>>> scalar_tensor = Tensor.one()
```

```
>>> value = scalar_tensor.scalar()
```

__iter__

Kind: Method

def __iter__(self) -> typing.Iterator[typing.Any]:

Iterator

Exhaustive reference · Source

TensorIndices

A tensor structure with abstract indices for symbolic tensor operations.

A tensor structure with abstract indices for symbolic tensor operations.

TensorIndices represents the index structure of tensors with named abstract indices that can be contracted, manipulated symbolically, and converted to expressions.

It maintains both the representation structure and index assignments.

Examples

```
>>> from symbolica.community.spenso import TensorIndices, Representation, TensorName
>>> rep = Representation.euc(3)
>>> mu = rep('mu')
>>> nu = rep('nu')
>>> indices = TensorIndices(mu, nu)
>>> T = TensorName("T")
>>> named_indices = T(mu, nu)
>>> expr = named_indices.to_expression()

class TensorIndices:
```

Requires features: python_stubgen

Inspect the registered export

```
from symbolica.community.spenso import TensorIndices
help(TensorIndices)
```

set_name

Kind: Method

```
def set_name(self, name: TensorName | builtins.str | Expression) -> None:
    Set the tensor name for this structure.
```

Parameters

name : TensorName

The tensor name to assign

Examples

```
>>> from symbolica.community.spenso import TensorIndices, TensorName, Representation
>>> rep = Representation.euc(3)
>>> structure = TensorIndices(rep('mu'), rep('nu'))
>>> T = TensorName("T")
>>> structure.set_name(T)
```

name

Kind: Parameter

name: TensorName | builtins.str | Expression

The tensor name to assign

get_name

Kind: Method

```
def get_name(self) -> typing.Optional[TensorName]:
    Get the tensor name of this structure.
```

Returns

TensorName or None

The tensor name if set, None otherwise

Examples

```
>>> name = structure.get_name()
```

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

__str__

Kind: Method

```
def __str__(self) -> builtins.str:
```

to_expression

Kind: Method

```
def to_expression(self) -> Expression:
```

Convert the tensor indices to a symbolic expression.

Creates a symbolic representation of the tensor with its indices that can be used in algebraic manipulations and pattern matching.

Returns

Expression

A symbolic Expression representing this indexed tensor

Raises

RuntimeError

If the tensor structure has no name

Examples

```
>>> from symbolica.community.spenso import TensorName, Representation
```

```
>>> T = TensorName("T")
```

```
>>> rep = Representation.euc(3)
```

```
>>> mu = rep('mu')
```

```
>>> nu = rep('nu')
```

```
>>> indices = T(mu, nu)
```

```
>>> expr = indices.to_expression()
```

__len__

Kind: Method

```
def __len__(self) -> builtins.int:
```

__add__

Kind: Method

```
def __add__(self, rhs: Expression | int | str | float | builtins.complex |  
TensorIndices | Expression) -> Expression:
```

Add this expression to `other`, returning the result.

rhs

Kind: Parameter

rhs: Expression | int | str | float | builtins.complex | TensorIndices | Expression

`__radd__`

Kind: Method

```
def __radd__(self, rhs: Expression | int | str | float | builtins.complex |  
TensorIndices | Expression) -> Expression:
```

Add this expression to `other`, returning the result.

rhs

Kind: Parameter

rhs: Expression | int | str | float | builtins.complex | TensorIndices | Expression

`__sub__`

Kind: Method

```
def __sub__(self, rhs: Expression | int | str | float | builtins.complex |  
TensorIndices | Expression) -> Expression:
```

Subtract `other` from this expression, returning the result.

rhs

Kind: Parameter

rhs: Expression | int | str | float | builtins.complex | TensorIndices | Expression

`__rsub__`

Kind: Method

```
def __rsub__(self, rhs: Expression | int | str | float | builtins.complex |  
TensorIndices | Expression) -> Expression:
```

Subtract this expression from `other`, returning the result.

rhs

Kind: Parameter

rhs: Expression | int | str | float | builtins.complex | TensorIndices | Expression

`__mul__`

Kind: Method

```
def __mul__(self, rhs: Expression | int | str | float | builtins.complex |  
TensorIndices | Expression) -> Expression:
```

Add this expression to `other`, returning the result.

rhs

Kind: Parameter

rhs: Expression | int | str | float | builtins.complex | TensorIndices | Expression

`__rmul__`

Kind: Method

```
def __rmul__(self, rhs: Expression | int | str | float | builtins.complex |  
TensorIndices | Expression) -> Expression:
```

Add this expression to `other`, returning the result.

rhs

Kind: Parameter

rhs: Expression | int | str | float | builtins.complex | TensorIndices | Expression

__new__

Kind: AssociatedFunction

```
def __new__(cls, *slots: TensorIndices | builtins.list[Slot], name: TensorName |
builtins.str | Expression | None = None) -> TensorIndices:
```

Create tensor structure from slots and optional arguments.

Parameters

*additional_args : Slot or Expression

Mixed arguments (Slot objects and Expressions for additional arguments)

name : TensorName, optional

Optional tensor name to assign to the structure

Returns

TensorIndices

A new TensorIndices object

Examples

```
>>> from symbolica import S
```

```
>>> from symbolica.community.spenso import TensorIndices, Representation, TensorName
```

```
>>> rep = Representation.euc(3)
```

```
>>> mu = rep('mu')
```

```
>>> nu = rep('nu')
```

```
>>> structure = TensorIndices(mu, nu)
```

```
>>> x = S('x')
```

```
>>> structure_with_args = TensorIndices(mu, nu, x)
```

```
>>> T = TensorName("T")
```

```
>>> named_structure = TensorIndices(mu, nu, name=T)
```

slots

Kind: Parameter

*slots: TensorIndices | builtins.list[Slot]

name

Kind: Parameter

name: TensorName | builtins.str | Expression | None = None

Default: None

Optional tensor name to assign to the structure

__getitem__

Kind: Method

```
def __getitem__(self, item: builtins.slice) -> builtins.list[Expression |
builtins.complex | float]:
```

Get expanded indices at the specified range of flattened indices.

Parameters

item : slice

Slice object defining the range of indices

Returns

list of list of int

List of expanded indices

item

Kind: Parameter

item: builtins.slice

Slice object defining the range of indices

__getitem__

Kind: Method

```
def __getitem__(self, item: typing.Sequence[builtins.int]) -> Expression |  
builtins.complex | float:
```

Get flattened index associated to this expanded index.

Parameters

item : list of int

Multi-dimensional index coordinates

Returns

int

The flat index

item

Kind: Parameter

item: typing.Sequence[builtins.int]

Multi-dimensional index coordinates

__getitem__

Kind: Method

```
def __getitem__(self, item: builtins.int) -> Expression | builtins.complex | float:
```

Get expanded index associated to this flat index.

Parameters

item : int

Flat index into the tensor

Returns

list of int

Multi-dimensional index coordinates

item

Kind: Parameter

item: builtins.int

Flat index into the tensor

Exhaustive reference · Source

TensorLibrary

A library for registering and managing tensor templates and structures.

A library for registering and managing tensor templates and structures.

The TensorLibrary provides a centralized registry for tensor definitions that can be reused across tensor networks and expressions. It manages tensor structures with their associated names and can resolve symbolic references to registered tensors.

```
```python
import symbolica
from symbolica.community.spenso import TensorLibrary, LibraryTensor, TensorStructure,
Representation

lib = TensorLibrary()
rep = Representation.euc(3)
name = symbolica.S("my_tensor")
structure = TensorStructure(rep, rep, name=name)
tensor = LibraryTensor.dense(structure, [1.0, 0.0, 0.0, 1.0, 0.0, 0.0, 1.0, 0.0,
0.0])
lib.register(tensor)
tensor_ref = lib[name]
```
```

```
class TensorLibrary:
```

Requires features: python_stubgen

Inspect the registered export

```
from symbolica.community.spenso import TensorLibrary
help(TensorLibrary)
```

__new__

Kind: AssociatedFunction

```
def __new__(cls) -> TensorLibrary:
```

Create a new empty tensor library.

Initializes an empty library ready for registering tensor structures.
The library automatically manages internal tensor IDs.

Returns

TensorLibrary

A new empty tensor library

Examples

```
>>> from symbolica.community.spenso import TensorLibrary
>>> lib = TensorLibrary()
```

construct

Kind: AssociatedFunction

```
def construct() -> TensorLibrary:
```

Create a new empty tensor library (static method).

Initializes an empty library ready for registering tensor structures.
The library automatically manages internal tensor IDs.

Returns

TensorLibrary

A new empty tensor library

Examples

```
>>> from symbolica.community.spenso import TensorLibrary
>>> lib = TensorLibrary.construct()
```

register

Kind: Method

```
def register(self, tensor: Tensor | LibraryTensor) -> None:
```

Register a tensor in the library.

Adds a tensor template to the library that can be referenced by name
in tensor networks and symbolic expressions. The tensor must have a name
set in its structure.

Parameters

tensor : LibraryTensor or Tensor

The tensor to register - can be a LibraryTensor or regular Tensor

Examples

```
>>> import symbolica
>>> from symbolica.community.spenso import TensorLibrary, LibraryTensor,
TensorStructure, Representation
>>> lib = TensorLibrary()
>>> rep = Representation.euc(3)
>>> name = symbolica.S("my_tensor")
>>> structure = TensorStructure(rep, rep, name=name)
>>> tensor = LibraryTensor.dense(structure, [1.0, 0.0, 0.0, 1.0, 0.0, 0.0, 1.0, 0.0,
0.0])
>>> lib.register(tensor)
>>> tensor_ref = lib[name]
```

tensor

Kind: Parameter

tensor: Tensor | LibraryTensor

The tensor to register - can be a LibraryTensor or regular Tensor

__getitem__

Kind: Method

```
def __getitem__(self, key: Expression | int | str | float | builtins.complex |
builtins.str) -> TensorStructure:
```

Retrieve a registered tensor structure by name.

Looks up a previously registered tensor by its name and returns a reference structure that can be used to create new tensor instances.

Parameters

key : str or Expression

The tensor name - can be a string or symbolic expression

Returns

TensorStructure

A TensorStructure representing the registered tensor template

Raises

RuntimeError

If the tensor name is not found in the library

Examples

```
>>> structure = lib["T"]
```

key

Kind: Parameter

key: Expression | int | str | float | builtins.complex | builtins.str

The tensor name - can be a string or symbolic expression

hep_lib

Kind: AssociatedFunction

```
def hep_lib() -> TensorLibrary:
```

Create a library pre-loaded with High Energy Physics tensor definitions.

Returns a library containing standard HEP tensors such as gamma matrices, color generators, metric tensors, and other commonly used structures in particle physics calculations. They are floating point tensors with f64 precision.

Returns

TensorLibrary

A TensorLibrary pre-populated with HEP tensor definitions

Examples

```
>>> import symbolica
>>> from symbolica.community.spenso import TensorLibrary, TensorName
>>> hep_lib = TensorLibrary.hep_lib()
>>> gamma_structure = hep_lib[symbolica.S("spenso::gamma")]
```

hep_lib_atom

Kind: AssociatedFunction

```
def hep_lib_atom() -> TensorLibrary:
```

Create a library pre-loaded with High Energy Physics tensor definitions.

Returns a library containing standard HEP tensors such as gamma matrices, color generators, metric tensors, and other commonly used structures in particle physics calculations. They are tensors with atom numeric entries.

Returns

TensorLibrary

A TensorLibrary pre-populated with HEP tensor definitions

Examples

```
>>> import symbolica
>>> from symbolica.community.spenso import TensorLibrary, TensorName
>>> hep_lib = TensorLibrary.hep_lib_atom()
>>> gamma_structure = hep_lib[symbolica.S("spenso::gamma")]
```

Exhaustive reference · Source

TensorNetwork

A tensor network representing computational graphs of tensor operations.

A tensor network representing computational graphs of tensor operations.

A tensor network is a graph-based representation of tensor computations where nodes represent tensors and operations, edges represent tensor contractions and data flow, and the network can be optimized and executed to compute results.

Tensor networks are particularly useful for symbolic manipulation of complex tensor expressions, optimization of tensor contraction orders, efficient evaluation of large tensor computations, and physics calculations involving many-body systems.

Examples

```
>>> import symbolica as sp
>>> from symbolica.community.spenso import TensorNetwork, Tensor, TensorIndices
>>> x = sp.symbol('x')
>>> expr = x * sp.symbol('T')(sp.symbol('mu'), sp.symbol('nu'))
>>> network = TensorNetwork(expr)
>>> network.execute()
>>> result = network.result_tensor()
```

class TensorNetwork:

Requires features: python_stubgen

Inspect the registered export

```
from symbolica.community.spenso import TensorNetwork
help(TensorNetwork)
```

__new__

Kind: AssociatedFunction

```
def __new__(cls, expr: Expression | int | str | float | builtins.complex |
TensorIndices | Expression, library: typing.Optional[TensorLibrary] = None) ->
TensorNetwork:
```

Create a tensor network by parsing an arithmetic expression.

Parses symbolic expressions containing tensor operations and converts them into an optimizable computational graph representation.

Parameters

expr : ArithmeticStructure

The arithmetic expression or tensor structure to parse

library : TensorLibrary, optional

Optional tensor library for resolving named tensor references

Returns

TensorNetwork

A new TensorNetwork representing the parsed expression

expr

Kind: Parameter

expr: Expression | int | str | float | builtins.complex | TensorIndices | Expression

The arithmetic expression or tensor structure to parse

library

Kind: Parameter

library: typing.Optional[TensorLibrary] = None

Default: None

Optional tensor library for resolving named tensor references

one

Kind: AssociatedFunction

def one() -> TensorNetwork:

Create a tensor network representing the scalar value 1.

Returns

TensorNetwork

A TensorNetwork containing only the scalar 1

Examples

```
>>> from symbolica.community.spenso import TensorNetwork
```

```
>>> one_net = TensorNetwork.one()
```

```
>>> result = one_net.result_scalar()
```

bracket

Kind: AssociatedFunction

def bracket() -> Expression:

broadcast

Kind: AssociatedFunction

def broadcast(str: builtins.str) -> Expression:

str

Kind: Parameter

str: builtins.str

zero

Kind: AssociatedFunction

def zero() -> TensorNetwork:

Create a tensor network representing the scalar value 0.

Returns

TensorNetwork

A TensorNetwork containing only the scalar 0

Examples

```
>>> from symbolica.community.spenso import TensorNetwork
```

```
>>> zero_net = TensorNetwork.zero()
```

```
>>> result = zero_net.result_scalar()
```

replace

Kind: Method

```
def replace(self, pattern: Expression | int | str | float | builtins.complex, rhs:
Expression | int | str | float | builtins.complex | HeldExpression |
typing.Callable[[dict[Expression, Expression]], Expression] | int | float | complex |
decimal.Decimal, _cond: typing.Optional[PatternRestriction | Condition] = None,
non_greedy_wildcards: typing.Optional[typing.Sequence[Expression]] = None,
level_range: typing.Optional[tuple[builtins.int, typing.Optional[builtins.int]]] =
None, level_is_tree_depth: typing.Optional[builtins.bool] = None,
allow_new_wildcards_on_rhs: typing.Optional[builtins.bool] = None, rhs_cache_size:
typing.Optional[builtins.int] = None, repeat: typing.Optional[builtins.bool] = None)
-> TensorNetwork:
```

Replace patterns in the tensor network using symbolic pattern matching.

Applies pattern-based transformations to the network structure, allowing for symbolic simplifications, substitutions, and algebraic manipulations.

Parameters

pattern : Expression

The symbolic pattern to match against

rhs : Expression

The replacement expression or pattern

non_greedy_wildcards : list of Expression, optional

List of wildcard symbols to match non-greedily

level_range : tuple of int, optional

Tuple specifying depth range for pattern matching

level_is_tree_depth : bool, optional

Whether level refers to tree depth or expression depth

allow_new_wildcards_on_rhs : bool, optional

Allow new wildcards in replacement pattern

rhs_cache_size : int, optional

Size of cache for replacement pattern compilation

repeat : bool, optional
Whether to repeatedly apply the replacement until no more matches

Returns

TensorNetwork

A new TensorNetwork with the replacements applied

pattern

Kind: Parameter

pattern: Expression | int | str | float | builtins.complex

The symbolic pattern to match against

rhs

Kind: Parameter

rhs: Expression | int | str | float | builtins.complex | HeldExpression |
typing.Callable[[dict[Expression, Expression], Expression] | int | float | complex |
decimal.Decimal

The replacement expression or pattern

_cond

Kind: Parameter

_cond: typing.Optional[PatternRestriction | Condition] = None

Default: None

non_greedy_wildcards

Kind: Parameter

non_greedy_wildcards: typing.Optional[typing.Sequence[Expression]] = None

Default: None

List of wildcard symbols to match non-greedily

level_range

Kind: Parameter

level_range: typing.Optional[tuple[builtins.int, typing.Optional[builtins.int]]] =
None

Default: None

Tuple specifying depth range for pattern matching

level_is_tree_depth

Kind: Parameter

level_is_tree_depth: typing.Optional[builtins.bool] = None

Default: None

Whether level refers to tree depth or expression depth

allow_new_wildcards_on_rhs

Kind: Parameter

allow_new_wildcards_on_rhs: typing.Optional[builtins.bool] = None

Default: None

Allow new wildcards in replacement pattern

rhs_cache_size

Kind: Parameter

rhs_cache_size: typing.Optional[builtins.int] = None

Default: None

Size of cache for replacement pattern compilation

repeat

Kind: Parameter

repeat: typing.Optional[builtins.bool] = None

Default: None

Whether to repeatedly apply the replacement until no more matches

evaluate

Kind: Method

```
def evaluate(self, constants: typing.Mapping[Expression, builtins.float], functions:
typing.Mapping[Expression, typing.Any]) -> TensorNetwork:
```

Evaluate symbolic expressions in the network with numerical values.

Substitutes symbolic constants and functions with numerical values,
converting symbolic parts of the network to concrete numerical tensors.

Parameters

constants : dict

Dict mapping symbolic expressions to their numerical values

functions : dict

Dict mapping function symbols to Python callable objects

Returns

TensorNetwork

A new TensorNetwork with symbolic expressions evaluated

constants

Kind: Parameter

constants: typing.Mapping[Expression, builtins.float]

Dict mapping symbolic expressions to their numerical values

functions

Kind: Parameter

functions: typing.Mapping[Expression, typing.Any]

Dict mapping function symbols to Python callable objects

execute

Kind: Method

```
def execute(self, library: typing.Optional[TensorLibrary] = None, function_library:
None = None, n_steps: typing.Optional[builtins.int] = None, mode: ExecutionMode
= ...) -> None:
```

Execute the tensor network to perform tensor contractions and simplifications.

Processes the computational graph by executing tensor operations such as contractions, additions, and multiplications. The execution can be controlled by mode and step limits.

Parameters

library : TensorLibrary, optional
Optional tensor library for resolving tensor operations
function_library : None, optional
Reserved for an internally supplied function library
n_steps : int, optional
Maximum number of execution steps (None for complete execution)
mode : ExecutionMode, optional
Execution strategy (ExecutionMode.All, ExecutionMode.Scalar, or ExecutionMode.Single)

Examples

```
>>> from symbolica.community.spenso import TensorNetwork, ExecutionMode,
TensorLibrary
>>> network = TensorNetwork(some_expression)
>>> network.execute()
>>> network.execute(n_steps=5)
>>> network.execute(mode=ExecutionMode.Scalar)
>>> lib = TensorLibrary.hep_lib()
>>> network.execute(library=lib)
```

library

Kind: Parameter

library: typing.Optional[TensorLibrary] = None

Default: None

Optional tensor library for resolving tensor operations

function_library

Kind: Parameter

function_library: None = None

Default: None

Reserved for an internally supplied function library

n_steps

Kind: Parameter

n_steps: typing.Optional[builtins.int] = None

Default: None

Maximum number of execution steps (None for complete execution)

mode

Kind: Parameter

mode: ExecutionMode = ...

Default: ...

Execution strategy (ExecutionMode.All, ExecutionMode.Scalar, or ExecutionMode.Single)

result_tensor

Kind: Method

```
def result_tensor(self, library: typing.Optional[TensorLibrary] = None) -> Tensor:
```

Extract the final tensor result from the executed network.

After network execution, retrieves the computed tensor result. The network should be executed before calling this method.

Parameters

library : TensorLibrary, optional

Optional tensor library for resolving tensor structures

Returns

Tensor

The computed tensor result

Raises

RuntimeError

If the network execution resulted in an error

Examples

```
>>> from symbolica.community.spenso import TensorNetwork, TensorLibrary
>>> network = TensorNetwork(tensor_expression)
>>> network.execute()
>>> result = network.result_tensor()
>>> lib = TensorLibrary.hep_lib()
>>> result_with_lib = network.result_tensor(library=lib)
```

library

Kind: Parameter

library: typing.Optional[TensorLibrary] = None

Default: None

Optional tensor library for resolving tensor structures

result_scalar

Kind: Method

```
def result_scalar(self) -> Expression:
```

Extract the final scalar result from the executed network.

For networks that evaluate to scalar expressions, retrieves the computed

scalar value. The network should be executed before calling this method.

Returns

Expression

The computed scalar expression

Raises

RuntimeError

If the network execution resulted in an error

Examples

```
>>> from symbolica.community.spenso import TensorNetwork
>>> network = TensorNetwork(scalar_expression)
>>> network.execute()
>>> scalar_result = network.result_scalar()
```

`__str__`

Kind: Method

```
def __str__(self) -> builtins.str:
```

Return a string representation of the network structure.

Generates a DOT format representation of the computational graph that can be visualized using graphviz or similar tools.

`__add__`

Kind: Method

```
def __add__(self, rhs: Expression | int | str | float | builtins.complex |
TensorIndices | Expression | TensorNetwork | Tensor) -> TensorNetwork:
```

Add two tensor networks element-wise.

Parameters

rhs : TensorNetwork

The tensor network to add (right-hand side)

Returns

TensorNetwork

A new TensorNetwork representing the sum

Examples

```
>>> net1 = TensorNetwork(expr1)
>>> net2 = TensorNetwork(expr2)
>>> sum_net = net1 + net2
```

rhs

Kind: Parameter

rhs: Expression | int | str | float | builtins.complex | TensorIndices | Expression | TensorNetwork | Tensor

The tensor network to add (right-hand side)

`__radd__`

Kind: Method

```
def __radd__(self, rhs: Expression | int | str | float | builtins.complex |
TensorIndices | Expression | TensorNetwork | Tensor) -> TensorNetwork:
```

Add two tensor networks element-wise (right-hand addition).

`rhs`

Kind: Parameter

`rhs`: Expression | int | str | float | builtins.complex | TensorIndices | Expression | TensorNetwork | Tensor

`__sub__`

Kind: Method

```
def __sub__(self, rhs: Expression | int | str | float | builtins.complex |
TensorIndices | Expression | TensorNetwork | Tensor) -> TensorNetwork:
```

Subtract one tensor network from another element-wise.

Parameters

`rhs` : TensorNetwork

The tensor network to subtract (right-hand side)

Returns

TensorNetwork

A new TensorNetwork representing the difference

Examples

```
>>> net1 = TensorNetwork(expr1)
```

```
>>> net2 = TensorNetwork(expr2)
```

```
>>> diff_net = net1 - net2
```

`rhs`

Kind: Parameter

`rhs`: Expression | int | str | float | builtins.complex | TensorIndices | Expression | TensorNetwork | Tensor

The tensor network to subtract (right-hand side)

`__rsub__`

Kind: Method

```
def __rsub__(self, rhs: Expression | int | str | float | builtins.complex |
TensorIndices | Expression | TensorNetwork | Tensor) -> TensorNetwork:
```

Subtract one tensor network from another (right-hand subtraction).

`rhs`

Kind: Parameter

`rhs`: Expression | int | str | float | builtins.complex | TensorIndices | Expression | TensorNetwork | Tensor

__mul__

Kind: Method

```
def __mul__(self, rhs: Expression | int | str | float | builtins.complex |
TensorIndices | Expression | TensorNetwork | Tensor) -> TensorNetwork:
```

Multiply two tensor networks.

Parameters

rhs : TensorNetwork

The tensor network to multiply with (right-hand side)

Returns

TensorNetwork

A new TensorNetwork representing the product

Examples

```
>>> net1 = TensorNetwork(expr1)
```

```
>>> net2 = TensorNetwork(expr2)
```

```
>>> product_net = net1 * net2
```

rhs

Kind: Parameter

rhs: Expression | int | str | float | builtins.complex | TensorIndices | Expression | TensorNetwork | Tensor

The tensor network to multiply with (right-hand side)

__rmul__

Kind: Method

```
def __rmul__(self, rhs: Expression | int | str | float | builtins.complex |
TensorIndices | Expression | TensorNetwork | Tensor) -> TensorNetwork:
```

Multiply two tensor networks (right-hand multiplication).

rhs

Kind: Parameter

rhs: Expression | int | str | float | builtins.complex | TensorIndices | Expression | TensorNetwork | Tensor

[Exhaustive reference](#) · [Source](#)

TensorStructure

A tensor structure without abstract indices, defined purely by representations.

A tensor structure without abstract indices, defined purely by representations.

TensorStructure represents the shape and representation structure of tensors without specific index assignments. It's used for defining tensor templates in libraries and for creating indexless tensor computations.

Examples:

```
```python
```

```
from symbolica.community.spenso import TensorStructure, Representation, TensorName
```

```

Create from representations
rep = Representation.euc(3)
structure = TensorStructure(rep, rep) # 3x3 matrix structure

With name for library registration
T = TensorName("T")
named_structure = TensorStructure(rep, rep, name=T)

Use to create indexed tensor
indices = structure.index('mu', 'nu') # Assign specific indices

Create symbolic expression
expr = structure.symbolic('a', 'b') # T(a, b)
` ``

```

```
class TensorStructure:
```

**Requires features:** python\_stubgen

### Inspect the registered export

```

from symbolica.community.spenso import TensorStructure
help(TensorStructure)

```

**set\_name**

**Kind:** Method

```
def set_name(self, name: TensorName | builtins.str | Expression) -> None:
```

**name**

**Kind:** Parameter

```
name: TensorName | builtins.str | Expression
```

**get\_name**

**Kind:** Method

```
def get_name(self) -> typing.Optional[TensorName]:
```

**\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> builtins.str:
```

**\_\_str\_\_**

**Kind:** Method

```
def __str__(self) -> builtins.str:
```

**\_\_len\_\_**

**Kind:** Method

```
def __len__(self) -> builtins.int:
```

**\_\_new\_\_**

**Kind:** AssociatedFunction

```

def __new__(cls, *reps_and_additional_args: TensorIndices | builtins.list[Slot],
name: TensorName | builtins.str | Expression | None = None) -> TensorStructure:

```



Construct a new TensorStructure with the given representations.

Parameters

-----

\*reps\_and\_additional\_args : Representation or Expression

Mixed arguments (Representation objects and Expressions for additional arguments)

name : TensorName, optional

Optional tensor name to assign to the structure

Returns

-----

TensorStructure

A new TensorStructure object

Examples

-----

```
>>> from symbolica import S
```

```
>>> from symbolica.community.spenso import TensorStructure, Representation,
TensorName
```

```
>>> rep = Representation.euc(3)
```

```
>>> structure = TensorStructure(rep, rep)
```

```
>>> x = S('x')
```

```
>>> structure_with_args = TensorStructure(rep, rep, x)
```

```
>>> T = TensorName("T")
```

```
>>> named_structure = TensorStructure(rep, rep, name=T)
```

**reps\_and\_additional\_args**

**Kind:** Parameter

\*reps\_and\_additional\_args: TensorIndices | builtins.list[Slot]

**name**

**Kind:** Parameter

name: TensorName | builtins.str | Expression | None = None

**Default:** None

Optional tensor name to assign to the structure

**\_\_getitem\_\_**

**Kind:** Method

```
def __getitem__(self, item: builtins.slice) -> builtins.list[Expression |
builtins.complex | float]:
```

Get expanded indices at the specified range of flattened indices.

Parameters

-----

item : slice

Slice object defining the range of indices

Returns

-----

list of list of int

List of expanded indices

**item**

**Kind:** Parameter

item: builtins.slice

Slice object defining the range of indices

**\_\_getitem\_\_**

**Kind:** Method

```
def __getitem__(self, item: typing.Sequence[builtins.int]) -> Expression |
builtins.complex | float:
```

Get flattened index associated to this expanded index.

Parameters

-----

item : list of int

Multi-dimensional index coordinates

Returns

-----

int

The flat index

**item**

**Kind:** Parameter

item: typing.Sequence[builtins.int]

Multi-dimensional index coordinates

**\_\_getitem\_\_**

**Kind:** Method

```
def __getitem__(self, item: builtins.int) -> Expression | builtins.complex | float:
```

Get expanded index associated to this flat index.

Parameters

-----

item : int

Flat index into the tensor

Returns

-----

list of int

Multi-dimensional index coordinates

**item**

**Kind:** Parameter

item: builtins.int

Flat index into the tensor

**\_\_call\_\_**

**Kind:** Method

```
def __call__(self, *args: builtins.int | Expression | str, extra_args:
typing.Sequence[Expression | int | str | float | builtins.complex] = []) ->
Expression:
```

Convenience method for creating symbolic expressions.

This is a shorthand for calling ``symbolic(*args, extra_args=extra_args)``.  
Creates a symbolic Expression representing this tensor structure.

Parameters

-----

`*args` : int, str, Symbol, or Expression  
Positional arguments (indices and additional args)  
`extra_args` : list of Expression, optional  
Optional list of additional non-tensorial arguments

Returns

-----

Expression  
A symbolic Expression representing the tensor

Examples

-----

```
>>> structure = TensorStructure(rep, rep, name="T")
>>> expr = structure('mu', 'nu')
```

**args**

**Kind:** Parameter

`*args`: builtins.int | Expression | str

**extra\_args**

**Kind:** Parameter

`extra_args`: typing.Sequence[Expression | int | str | float | builtins.complex] = []

**Default:** []

Optional list of additional non-tensorial arguments

**symbolic**

**Kind:** Method

```
def symbolic(self, *args: builtins.int | Expression | str, extra_args:
typing.Sequence[Expression | int | str | float | builtins.complex] = []) ->
Expression:
```

Create a symbolic expression representing this tensor structure.

Builds a symbolic tensor expression with the specified indices. Arguments can be separated using a semicolon (';') to distinguish between additional arguments and tensor indices.

Parameters

-----

`*args` : int, str, Symbol, Expression, or ';'   
Positional arguments (int, str, Symbol, Expression for indices, ';' for separator)  
`extra_args` : list of Expression, optional  
Optional list of additional non-tensorial arguments

Returns

-----

Expression

A symbolic Expression representing the tensor with indices

Examples

-----

```
>>> import symbolica as sp
>>> from symbolica.community.spenso import TensorStructure, Representation,
TensorName
>>> rep = Representation.euc(3)
>>> T = TensorName("T")
>>> structure = TensorStructure([rep, rep], name=T)
>>> expr = structure.symbolic('mu', 'nu')
>>> x = sp.S('x')
>>> expr = structure.symbolic(x, ';', 'mu', 'nu')
>>> expr = structure.symbolic('mu', 'nu', extra_args=[x])
```

**args**

**Kind:** Parameter

\*args: builtins.int | Expression | str

**extra\_args**

**Kind:** Parameter

extra\_args: typing.Sequence[Expression | int | str | float | builtins.complex] = []

**Default:** []

Optional list of additional non-tensorial arguments

**index**

**Kind:** Method

```
def index(self, *args: builtins.int | Expression | str, extra_args:
typing.Sequence[Expression] = [], cook_indices: builtins.bool = False) ->
TensorIndices:
```

Create an indexed tensor (TensorIndices) from this structure.

Converts this structure template into a concrete indexed tensor by assigning specific abstract indices to each representation slot.

Parameters

-----

\*args : int, str, Symbol, Expression, or ';'

Positional arguments (indices and ';' separator between additional args and indices)

extra\_args : list of Expression, optional

Optional list of additional non-tensorial arguments

cook\_indices : bool, optional

If True, attempt to convert expressions to valid indices

Returns

-----

TensorIndices

A TensorIndices object with concrete index assignments

## Examples

-----

```
>>> import symbolica as sp
>>> from symbolica.community.spenso import TensorStructure, Representation,
TensorName
>>> rep = Representation.cof(3)
>>> T = TensorName("T")
>>> structure = TensorStructure([rep, rep], name=T)
>>> indices = structure.index('mu', 'nu')
>>> x = sp.S('x')
>>> indices = structure.index(x, ';', 'mu', 'nu')
```

## args

**Kind:** Parameter

\*args: builtins.int | Expression | str

## extra\_args

**Kind:** Parameter

extra\_args: typing.Sequence[Expression] = []

**Default:** []

Optional list of additional non-tensorial arguments

## cook\_indices

**Kind:** Parameter

cook\_indices: builtins.bool = False

**Default:** False

If True, attempt to convert expressions to valid indices

[Exhaustive reference](#) · [Source](#)

## ExecutionMode

Execution modes for tensor network evaluation.

Execution modes for tensor network evaluation.

Controls how the tensor network execution engine processes the computational graph.

## Variants

-----

Single : Execute one contraction at a time, useful for debugging

Scalar : Only contract scalar operations, leaving tensor structure intact

All : Execute all possible contractions for complete evaluation

```
class ExecutionMode(enum.Enum):
```

**Requires features:** python\_stubgen

## Inspect the registered export

```
from symbolica.community.spenso import ExecutionMode
help(ExecutionMode)
```

[Exhaustive reference](#) · [Source](#)

## SymbolicParallelism

Policy for Rayon operations that manipulate Symbolica expressions.

Policy for Rayon operations that manipulate Symbolica expressions.

```
class SymbolicParallelism(enum.Enum):
```

**Requires features:** python\_stubgen

## Inspect the registered export

```
from symbolica.community.spenso import SymbolicParallelism
help(SymbolicParallelism)
```

Exhaustive reference · Source

# Canonical package changelog

The following release history is imported as typed Markdown from crates/spenso/CHANGELOG.md.

## Changelog

All notable changes to this project will be documented in this file.

The format is based on Keep a Changelog, and this project adheres to Semantic Versioning.

### [Unreleased]

#### 0.5.6 - 2026-01-08

##### Fixed

- fix python\_stub\_gen feature
- fix metric initialization and update to symbolica 1.2

##### Other

- update linnet

#### 0.5.5 - 2025-12-15

##### Fixed

- fix canonization in presence of duals
- fix sum normalization in network parsing
- fix warnings
- fix overflowing slot order
- fix add\_assign with sparse tensors

##### Other

- Update symbolica to 1.1.0
- update linnet
- add depth to parser, to stop at first mul
- remove more prints
- remove prints
- update linnet
- add readmes
- formatting and add ref for parametric
- update to 0.17 pyo3\_stub\_gen
- update linnet and make classattr to classmethods
- Release spenso-macros 0.3
- update to symbolica 1.0
- update sympolica to 0.20
- Properly precontract scalars
- Add back pyo3-stub-gen-derive
- Updated spenso with minor canonical string update for symbolica f2eb0c1ddcc61c342f2ff9616c2d129141d433dc
- Refactor coefficient conversions to support generic float types
- update to current dev symbolica and add pattern for pslash + m conjugate
- update to linnet 0.13.1
- working canonize (up to functions) and dirac adjoint
- spenso canonize buggy

- robust\_conj but with lots of gamma\_0s
- first try conj
- update python api
- working pow and fun execution
- working pow execution with wrapping
- support for executing pow and functions on tensors
- implement function library trait
- add function generic key

### **0.5.3 - 2025-08-18**

#### **Other**

- update linnet to crates
- all tests pass
- Fix Multiple Heads Bug
- implement ExportNumber for spenso::Complex
- to Atom for ExpandedIndex
- remove printouts
- update linnet
- update linnet
- update linnet
- add infinite loop error test
- update parse problem and add spenso:: to all symbols
- remove expand

### **0.5.2 - 2025-07-15**

#### **Other**

- remove printout

### **0.5.1 - 2025-07-15**

#### **Fixed**

- fix gamma algebra≈

#### **Other**

- allow arbitrary arguments for to dots



## Version information

Save these identifiers with published results and include them in issue reports so that collaborators can reproduce the same software environment.

- **Documentation channel:** latest
- **Documentation route:** /products/spenso/latest/
- **Source revision:** dfddcec2a39e23d2d5faba16c18f197dc3301473

## Package versions and enabled features

- gammaloop/gammalooprs — 0.3.4 (no optional features)
- gammaloop/gammaloop-api — 0.1.0 (features: cli)
- gammaloop/gammaloop — 0.3.4 (features: python\_api, python\_abi, python\_stubgen)
- linnet/linnet — 0.17.0 (features: nodestore-vec, serde, bincode, drawing, symbolica)
- linnet/linnet-py — 0.1.0 (features: extension-module, abi3-py310)
- spenso/spenso — 0.6.0 (features: shadowing)
- spenso/spenso-macros — 0.3.2 (features: shadowing)
- spenso/spenso-hep-lib — 0.1.7 (no optional features)
- spenso/spynso3 — 0.1.2 (features: python\_stubgen)
- idenso/idenso — 0.3.0 (features: bincode, reference-cases)
- idenso/idenso — 0.3.0 (features: python, python\_stubgen)
- vakint/vakint — 0.1.2 (no optional features)
- vakint/vakint — 0.1.2 (features: symbolica\_community\_module, python\_stubgen)